PREPARED FOR TRIPLETEN · LEARNING EXPERIENCE LEAD · AUGUST 2026

Greg T. *Chism*

PH.D. · LEARNING EXPERIENCE PORTFOLIO

GREGTCHISM.COM · GITHUB · LINKEDIN · GCHISM94@GMAIL.COM

I design learning experiences end-to-end: identifying the opportunity, building the course and the platform it runs on, and measuring whether people learned. For the past four years I've done that in data science, machine learning, and AI education at the University of Arizona. My courses are built in public, revised every term, and designed with AI tools at every stage, from research and prototyping to content and feedback. I'm also a Carpentries Instructor Trainer, so I train other instructors in the same evidence-based methods I use.

6	100+	40+	*$368K*	8
GRADUATE COURSES DESIGNED & TAUGHT	CAPSTONE STUDENTS (GREW PROGRAM FROM 7)	GRAD STUDENTS & POSTDOCS MENTORED	NSF COMPUTING AWARDS WON AS LEAD-PI	PEER-REVIEWED ARTICLES

01 · COURSES

Designed end-to-end, shipped in public

Each course below is a complete educational product, from audience analysis through delivery and revision. Materials are published openly, so the design decisions can be inspected and reused.

CASE STUDY 01 — TEACHING BEGINNERS

INFO 511 · Fundamentals of Data Science

UNIVERSITY OF ARIZONA ISCHOOL · DATASCIAZ.NETLIFY.APP

THE BRIEF

Take graduate students, many arriving from non-technical backgrounds, through the entire data science lifecycle — collection, cleaning, exploration, modeling, communication — in a single semester.

THE DESIGN

Python-first with real-world datasets, so every skill is learned in the context it will be used. Reproducible workflows (Git, Quarto) from week one, and modular AI tutoring from my research — retrieval-augmented feedback, adaptive quizzes, and code-similarity checking.

THE EVIDENCE

The full course is open source, with automated GitHub Classroom test suites checking correctness, style, and edge cases — professionals enrolled confirm it matches how they work.

CASE STUDY 02 — DESIGNING WHERE NO PLAYBOOK EXISTS

INFO 557 · Neural Networks

UNIVERSITY OF ARIZONA ISCHOOL · NEURALNETWORKSAZ-F25.NETLIFY.APP · INTERACTIVE VIZ

THE BRIEF

Build a graduate deep-learning course in a domain that changes every semester, where proven instructional playbooks don't exist yet.

THE DESIGN

Fundamentals-first: backpropagation, optimization, and regularization built up from scratch, so learners have the foundations to follow each new architecture wave. Students build, train, and evaluate real models, with gamified exams and a custom interactive visualization that makes the invisible mechanics visible.

THE EVIDENCE

Versioned by semester, and powered by an AI Sandbox I fund as Lead-PI of a NAIRR Classroom Pilot — the university's only competitively awarded NAIRR Pilot allocation.

CASE STUDY 03 — BUSINESS GOALS → LEARNING OUTCOMES

INFO 698 · Data Science Capstone

UNIVERSITY OF ARIZONA ISCHOOL · INDUSTRY-SPONSORED PROJECTS

THE BRIEF

Industry sponsors arrive with business problems. The design task is turning them into scoped, achievable capstones that serve both the sponsor and the learner.

THE DESIGN

I help scope sponsor goals into well-defined projects, then support student teams from scoping through delivery with mentored check-ins and free instructional computing resources.

THE EVIDENCE

Grew the program from seven students to 100+ — the College's largest applied program — and a $245.5K NSF award I won as Lead-PI let 10+ teams build LLM projects a standard course couldn't run.

INFO 521 · Intro to Machine Learning — ML from the mathematics up, implemented from scratch in Python rather than off-the-shelf libraries.

INFO 523 · Data Mining & Discovery — knowledge discovery with hands-on, reproducible workflows. DATAMINEAZ.ORG

INFO 526 · Data Analysis & Visualization — graphic design, programming, and statistics for publication-ready figures in R and Python. DATAVIZAZ.ORG

02 · AI-FIRST LEARNING ENGINEERING

The platforms behind the courses

I use AI throughout my design work. Prompt engineering, the Claude and OpenAI APIs, RAG, and agentic workflows are everyday tools for research, prototyping, content production, and feedback.

NAIRR AI Sandbox

LEAD-PI · 30K GPU CREDITS (~$50K) · NSF NAIRR · 2025

A national-infrastructure sandbox for applied neural-network education, where learners experiment on real GPUs. I designed the sandbox and won the award as Lead-PI.

Sol — Agentic Tutoring & Feedback

UNIVERSITY OF ARIZONA ISCHOOL · 2023-PRESENT

Sol is an open-source agentic AI with modular tutoring and feedback components, built from my research and piloted across my courses to scale support beyond instructor hours.

JupyterQuest

OPEN SOURCE · ZENODO DOI · GITHUB.COM/GCHISM94

A gamified, interactive Jupyter environment for teaching reproducible data science — motivation design and automated checks built directly into the learner's working environment.

MESA, AI Core & EduCloud

NSF EDUCATION LEAD · FACULTY DIRECTOR · 2025-PRESENT

Lead education & outreach for MESA, a $4.6M NSF cyberinfrastructure project; supervise an AI + XR/VR experiential program; building EduCloud, provider-agnostic educational computing.

03 · MEASUREMENT

Measuring whether it worked

MEASUREMENT SCIENCE	I publish peer-reviewed research on whether instruments measure what they claim (*J. Affective Disorders*, 2026, on PHQ-8 reliability). I apply the same standards when defining learning success metrics.
INSTRUMENTED ITERATION	Course repositories are versioned and public. **Every term's revision is traceable to what learners did**, and the changes stay reviewable.
RESEARCH-GRADE RIGOR	Ph.D. training in empirical research design; 8 peer-reviewed articles; reproducible analysis pipelines built with Binder and GitHub Actions.

04 · MENTORING & STANDARDS

Training and mentoring other educators

INSTRUCTOR TRAINER	**Carpentries Instructor Trainer (2023–present)** — I train and certify new instructors worldwide in evidence-based, inclusive teaching practice.
MENTORSHIP AT SCALE	Mentored **40+ graduate students and postdocs** and 24 capstone team projects; primary PhD advisor for dissertation research on AI sandboxes in education.
SCALABLE STANDARDS	Open, reusable course materials; chaired Research Bazaar Arizona, expanding it across three universities.